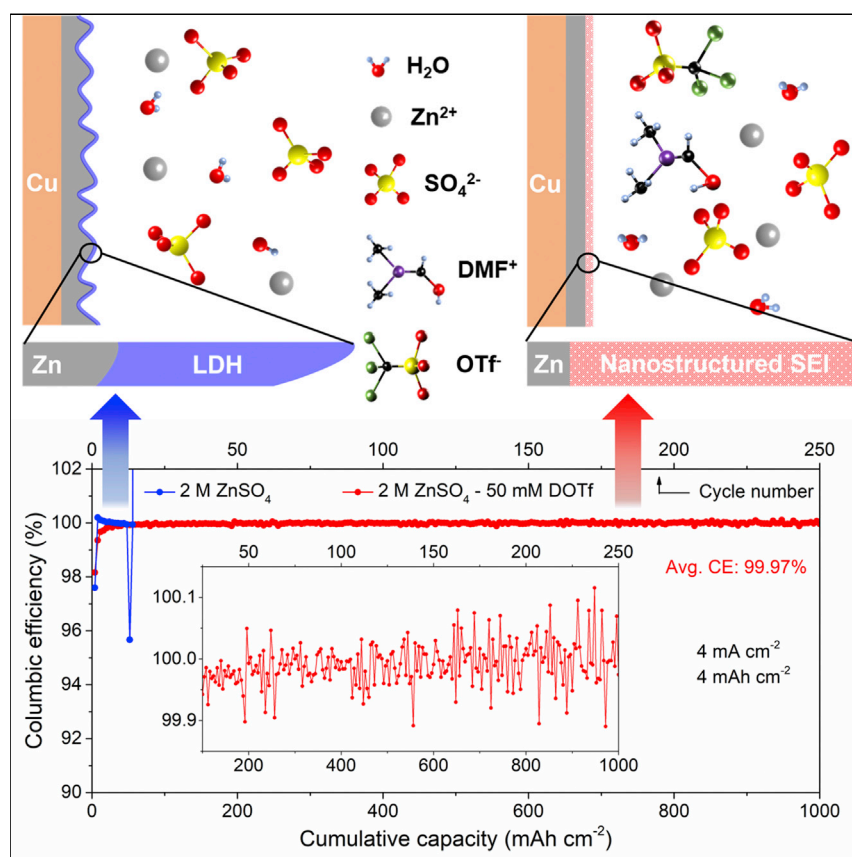


Article

Highly reversible Zn anode with a practical areal capacity enabled by a sustainable electrolyte and superacid interfacial chemistry



Tremendous effort has been devoted to exploring aqueous zinc-metal batteries for stationary electrochemical energy storage. However, the poor reversibility and stability of Zn anodes has plagued their commercialization. This work introduces a novel electrolyte additive that effectively enhances Zn anode performance in conventional aqueous systems to fulfill practical requirements. The superacid interfacial chemistry and formation of a nanostructured solid-electrolyte interface induce nanoscale nucleation/growth and dense Zn deposition.

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Highlights

DOTf was introduced as an additive for low-cost aqueous Zn electrolytes

Highly reversible Zn plating/stripping is achieved under practical conditions

Key is dissociation of DOTf to HOTf that assists in forming a nanostructured SEI

Full cell shows thousands of cycles with a mass-limited Zn anode

Article

Highly reversible Zn anode with a practical areal capacity enabled by a sustainable electrolyte and superacid interfacial chemistry

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SUMMARY

Aqueous zinc-metal batteries are plagued by poor Zn reversibility owing to zinc dendrite and layered double hydroxide (LDH) formation. Here, we introduce a novel additive—N,N-dimethylformamidium trifluoromethanesulfonate (DOTf)—in a low-cost aqueous electrolyte that can very effectively address these issues. The initial water-assisted dissociation of DOTf into triflic superacid creates a robust nanostructured solid-electrolyte interface (SEI)—revealed by *operando* spectroscopy and cryomicroscopy—which excludes water and enables dense Zn deposition. We demonstrate excellent Zn plating/stripping in a Zn||Cu asymmetric cell for more than 3,500 cycles. Furthermore, near 100% CE is realized at a combined high current density of 4 mA cm⁻² and an areal capacity of 4 mAh cm⁻² over long-term cycling. Zn||Zn_{0.25}V₂O₅·nH₂O full cells retain ~83% of their capacity after 1,000 cycles with mass-limited Zn anodes. By restricting the depth of discharge, the cathodes exhibit less proton intercalation and LDH formation with an extended lifetime of 2,000 cycles.

INTRODUCTION

Aqueous zinc-metal batteries (AZMBs) are one of most promising alternative energy storage technologies to Li-ion batteries (LIBs).^{1,2} They benefit from the high abundance and large theoretical capacity of Zn (5,851 mAh mL⁻¹ and 820 mAh g⁻¹), as well as superior safety ascribed to their high compatibility with non-flammable aqueous electrolytes.^{1,2} The resulting affordability and safety of AZMBs are particularly appealing for grid or mini-grid stationary storage.^{1–3} Many cathode materials for this type of battery exhibit higher specific capacity and superior cycling stability than those reported for LIBs, including MnO₂⁴ and Zn_{0.25}V₂O₅·nH₂O.⁵ However, the poor reversibility of the Zn metal anode has impeded commercialization of AZMBs. Owing to the high activity of H₂O, triggering of the hydrogen evolution reaction (HER) during Zn deposition leads to a local alkaline environment in common aqueous electrolytes (for example, 2 M ZnSO₄).⁶ This induces the formation of high surface area zinc layered double hydroxides (LDHs) such as Zn₄SO₄(OH)₆·xH₂O, and zinc hydroxide on the Zn surface, resulting in a non-homogeneous Zn surface that leads to low Coulombic efficiency (CE) of Zn plating/stripping, aggravated dendrite formation and accelerated battery failure.^{6–8}

To overcome these challenges and improve the reversibility of Zn, various intriguing strategies have been employed, including building three-dimensional (3D) Zn hosts or employing Zn/substrate surface modification.^{9–14} A 3D metal-organic framework

Context & scale

Aqueous zinc-metal batteries are attractive for stationary electrochemical energy storage; however, the low-density growth habit of Zn deposited at the anode in conventional, slightly acidic aqueous electrolytes greatly limits cycling lifetimes. In this report, practical-level Zn anode performance is achieved in a low-cost 2 M ZnSO₄ aqueous electrolyte by utilizing a novel additive: N,N-dimethylformamidium trifluoromethanesulfonate (DOTf). Advanced *operando* spectroscopy and cryomicroscopy reveal that the water-assisted dissociation of DOTf to triflic acid forms a nanostructured SEI on the deposited Zn surface. This robust SEI suppresses the formation of deleterious side products and leads to uniform nucleation of Zn metal nanocrystallites, forming dense and highly compact Zn films with a dendrite-free morphology. Zn||Zn_{0.25}V₂O₅·nH₂O full cells exhibit stable behavior for thousands of cycles under rigorous conditions with a mass-limited Zn anode.



(ZIF-8) exhibits a high Zn plating/stripping CE of 99% for 200 cycles at a range of current densities and capacities,⁹ although scalability of ZIF synthesis is a consideration for large-scale AZMBs. Zheng et al. reported an epitaxial mechanism to regulate Zn nucleation and growth. Owing to the low lattice mismatch between Zn and graphene, a remarkable performance based on dendrite-free Zn plating/stripping on a graphene substrate was achieved via “locked orientation.”¹⁵ A similar approach was extended to electrolyte-dispersed C₃N₄ colloidal particles, which improve Zn electrodeposition to achieve a CE of 99.8%.¹⁶ Electrolyte engineering via tuning the interaction of cation, anion, and solvent is another effective approach to enhance Zn reversibility and stability.^{17,18} Wang’s group reported a neutral 21 m LiTFSI + 1 m Zn(TFSI)₂ water-in-salt electrolyte (WiSE) in which H₂O molecules are strongly crowded by Li⁺ ions, thus exhibiting reduced H₂O activity and a high CE of 99.7%.¹⁷ Nonetheless, very significant polarization (around 0.3 V at 0.2 mA cm⁻²) and high cost of the salt are potential limitations of the WiSE systems.

Recently, ZnF₂-rich solid-electrolyte interfaces (SEIs) formed *in situ* on Zn metal have been proposed to enhance Zn reversibility. Cao et al. reported a metal-organic framework layer filled with a Zn(TFSI)₂-tris(2,2,2-trifluoroethyl)-phosphate organic electrolyte.¹⁹ Zn anodes coated with this layer show an average CE of 99.1% for 350 cycles in 1 M Zn(TFSI)₂ aqueous electrolyte, where a ZnF₂-rich SEI *in situ* formed on Zn surface was proposed to enhance the Zn reversibility. Wang’s group further explored the effect of trimethylethylammonium trifluoromethanesulfonate (Me₃EtNOTf) as an additive with 4 m Zn(OTf)₂, reporting an average CE of 99.8% for 1,000 cycles.²⁰ A ZnF₂-SEI formed *in situ* on the Zn surface was proposed to explain such good performance, operating at a current density of 0.5 mA.cm⁻². Inspired by the promising properties of a ZnF₂-rich SEI, Han et al. designed an artificial porous ZnF₂ layer on Zn with a thickness of ~1 μm that effectively suppressed dendrite growth.²¹ Their modeling results remarkably showed that Zn²⁺ could diffuse through bulk ZnF₂ as a result of Zn²⁺-interstitial defect formation. Zn²⁺-ion migration via the channels embedded in this fluoride layer may also play a role. Hybrid solvents with minimal water content are also proposed to suppress water activity and enhance Zn reversibility.^{22,23} Very recently, Han et al. have reported a hydrous organic electrolyte prepared by dissolving hydrated Zn(BF₄)₂ in ethylene glycol.²³ This electrolyte enables the formation of a dense ~3 μm ZnF₂ passivation layer that exhibits a relatively high CE (99.4%) for Zn plating/stripping. Despite the high Zn reversibility owing to the ZnF₂-rich SEI, the low areal capacity (0.5 mAh cm⁻²) achieved in these studies is less than that of practical LIBs (at least 3 mAh cm⁻²).²⁴

Herein, we introduce a novel electrolyte additive, N,N-dimethylformamidium trifluoromethanesulfonate (denoted as DOTf), for an electrolyte based on aqueous 2 M ZnSO₄. Its dissociation in water to neutral ion pairs (the superacid triflic acid [HOTf] + N,N-dimethylformamide [DMF]) leads to the formation of a fluoride-free, Zn²⁺-conducting nanostructured SEI on the Zn surface upon cycling in an electrochemical cell. The unique interfacial chemistry is understood by using a comprehensive combination of X-ray photoelectron spectroscopy (XPS), *operando* surface-enhanced infrared absorption spectroscopy (SEIRAS), and cryo-transmission electron microscopy (TEM) studies. We demonstrate that the SEI enables water exclusion at the interface, leading to diminished LDH formation. Along with a nano-scale Zn nucleation and growth process, this additive promotes a highly dense and dendrite-free Zn-plating morphology. The electrochemistry of Zn||Cu asymmetric cells shows highly stable and reversible Zn plating/stripping at both a lab-level capacity of 0.5 mAh cm⁻² and a practical-level capacity of 4 mAh cm⁻². More importantly, this electrolyte is highly compatible with Zn||Zn_{0.25}V₂O₅·nH₂O full cells at a

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<https://doi.org/10.1016/j.joule.2022.04.017>

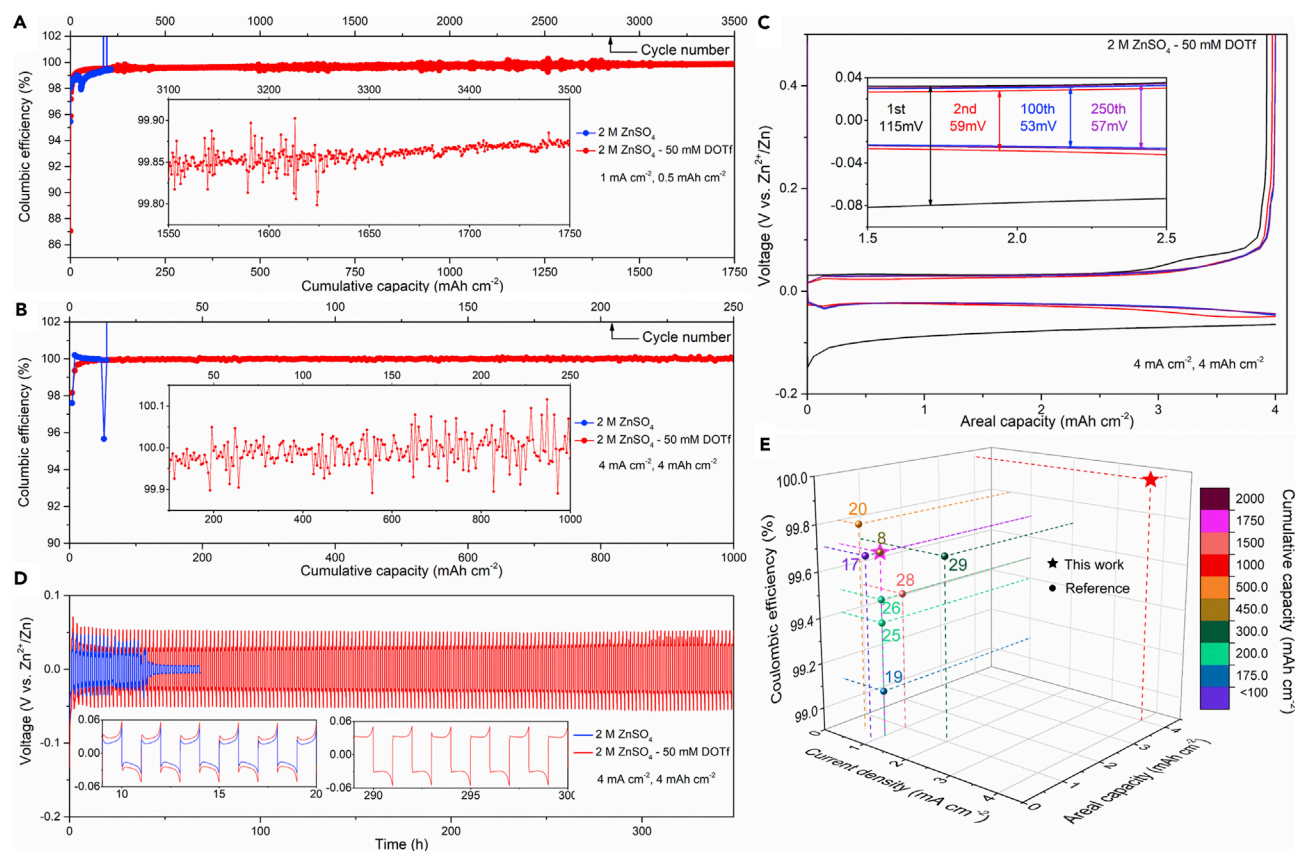


Figure 1. Electrochemistry studies on Zn plating/stripping in different electrolytes

(A and B) CE evolution on cycling Zn||Cu asymmetric cells under different conditions: (A) 1 mA cm^{-2} and 0.5 mAh cm^{-2} and (B) 4 mA cm^{-2} and 4 mAh cm^{-2} . Slight variation ($\pm 0.1\%$) in CE is to be expected.

(C) Voltage profiles of the Zn||Cu cell in 2 M ZnSO_4 - 50 mM DOTf at 4 mA cm^{-2} and 4 mAh cm^{-2} .

(D) Galvanostatic Zn stripping/plating in Zn||Zn symmetric cells at 4 mA cm^{-2} and 4 mAh cm^{-2} .

(E) Comparison between the Zn plating/stripping performance of asymmetric cells in this work and that reported based on four critical parameters: Coulombic efficiency, current density, areal capacity, and cumulative capacity. Only electrolytes that have demonstrated an average CE of $>99\%$ for at least 200 cycles are selected (see Table S1).

low negative: positive (N:P) electrode ratio of 2.5, exhibiting a capacity retention of $\sim 83\%$ after 1,000 cycles. By limiting the depth of discharge, an enhanced life of more than 2,000 cycles can be achieved with less proton intercalation.

RESULTS AND DISCUSSION

Highly reversible Zn plating/stripping

The Zn plating/stripping behavior was examined in Zn||Cu asymmetric cells at 1 mA cm^{-2} and 0.5 mAh cm^{-2} —a frequently reported CE test condition for Zn anodes.^{8,19,23,25,26} As shown in Figures 1A and S1, the CE in 2 M ZnSO_4 - 50 mM DOTf quickly increases to 99% in the initial 20 cycles and finally stabilizes at $>99.8\%$ for 3,500 cycles (Figure S2 and R1). To the best of our knowledge, this is the longest cycle life and highest cumulative capacity achieved under these conditions of current density and areal capacity. This results in a high average CE value of 99.7% for a total cumulative capacity of $1,750 \text{ mAh cm}^{-2}$. In the 2 M ZnSO_4 baseline electrolyte (Figure 1A), the Zn||Cu cell shows a fluctuating CE and fails after 214 cycles. More importantly, at practical conditions of 4 mA cm^{-2} and 4 mAh cm^{-2} , the CE in the modified electrolyte reaches over 99.9% within 10 cycles and is preserved for 250 cycles (Figure 1B and R2). This accounts for an unprecedented high average

CE of 99.97% for a total $1,000 \text{ mAh cm}^{-2}$ cumulative capacity. The voltage profiles in Figure 1C validate the highly reversible Zn plating/stripping at such high current density and capacity in the presence of DOTf, as confirmed by the small and stable overpotential of $\sim 28 \text{ mV}$ after the initial cycle in contrast to the fluctuating plating/stripping curves in the baseline electrolyte (Figure S3). In addition, at even more aggressive conditions of 10 mA cm^{-2} and 10 mAh cm^{-2} , a high average CE of 99.8% can still be achieved (Figure S4 and R3). Figure 1D displays the excellent stability of a Zn||Zn symmetric cell with 2 M ZnSO_4 - 50 mM DOTf , showing steady cycling at 4 mA cm^{-2} and 4 mAh cm^{-2} for nearly 350 h. This is approximately one order of magnitude longer than that without DOTf, where the cell shorts and quickly fails. We ascribe the slightly higher cell polarization in the presence of the additive ($\sim 5 \text{ mV}$) to *in situ* formation of a passivating SEI, as reported elsewhere²⁰ and discussed below. We note that the “soft shorts” widely reported in symmetric cell testing²⁷ are not observed here, as confirmed by the stable and non-zero charge transfer resistance throughout the cycling (Figure S5).

Four important parameters: Zn plating/stripping CE, current density, areal capacity of each plating and stripping, as well as the total cumulative capacity were selected to compare our modified electrolyte with other reported Zn electrolytes.^{8,17,19,20,25,26,28,29} As shown in Figure 1E and Table S1, while the metrics exhibited in this work are comparable to, or better than other electrolytes at similar lab-level conditions (in our case, 1 mA cm^{-2} and 0.5 mAh cm^{-2}), we also demonstrate excellent performance under practical conditions (4 mA cm^{-2} and 4 mAh cm^{-2}) with a high CE of 99.97%. It is critical to screen Zn reversibility at both high and low current densities to fully investigate its commercial viability.^{2,30} Such promising enhancement of Zn plating/stripping performance can also be extended to other electrolyte systems (Figure S6).

Dendrite-free Zn-plating morphology

Cu electrodes with plated Zn, retrieved from the asymmetric cells after the 10th plating at 4 mA cm^{-2} and 4 mAh cm^{-2} were analyzed by scanning electron microscopy (SEM). Zn plated in 2 M ZnSO_4 - 50 mM DOTf shows a smooth and compact layer, formed by merged homogeneous Zn particles with dimensions of $\sim 250 \text{ nm}$ (Figures 2A and 2B), whereas porous and dendritic Zn was observed in 2 M ZnSO_4 as confirmed by energy-dispersive X-ray spectroscopy (EDX, Figure S7), consistent with the rapid battery failure in both Zn||Cu and Zn||Zn cells (Figures 2E and 2F). The cross-sectional view of deposited Zn shows a spongy morphology with a thickness of $\sim 13 \mu\text{m}$ in 2 M ZnSO_4 (Figures 2G, 2H, and S8), whereas a thin $\sim 6 \mu\text{m}$ layer is observed in the modified electrolyte (Figures 2C, 2D, and S9). This is nearly six times denser than the reported Zn deposition layer with a fluorinated SEI,²⁰ based on calculation of the Zn capacity and thickness described in Cao et al.²⁰ Similar morphologies in both electrolytes were also confirmed at 1 mA cm^{-2} and 0.5 mAh cm^{-2} (Figure S10).

These results show the effectiveness of the DOTf additive to suppress dendrite growth. A combination of nuclear magnetic resonance (NMR) and XPS studies described below indicate that this can be ascribed to a unique Zn^{2+} -conducting SEI formed on the Zn surface.

Forming the Zn^{2+} -conducting SEI

The ^1H NMR spectrum of the 2 M ZnSO_4 - 50 mM DOTf solution (Figure S11) shows characteristic peaks of DMF at $\delta = 2.42, 2.57, \text{ and } 7.49 \text{ ppm}$, along with two additional peaks at $\delta = 2.28 \text{ and } 7.85 \text{ ppm}$, which are not observed in the spectrum of

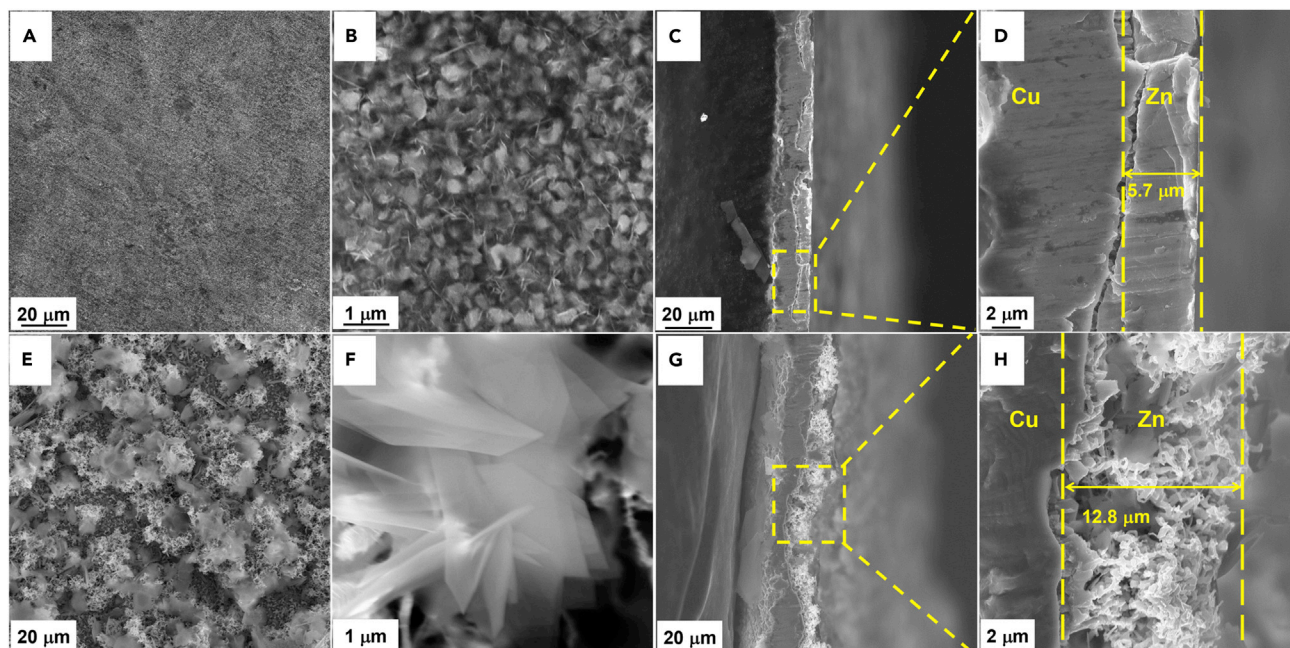


Figure 2. Morphologies of plated Zn on Cu after 10 cycles at 4 mA cm^{-2} and 4 mAh cm^{-2}

(A–D) SEM images of plated Zn in 2 M ZnSO_4 -50 mM DOTf: (A and B) top view and (C and D) cross-sectional view.

(E–H) SEM images of plated Zn in 2 M ZnSO_4 : (E and F) top view and (G and H) cross-sectional view. Images shown and indicated thicknesses are representative of the average deposition layers.

2 M ZnSO_4 -50 mM DMF. Upon adding 50 mM HOTf into a 2 M ZnSO_4 -50 mM DMF solution, the two additional peaks appear and lead to an identical pattern (Figure 3A). These results indicate that DOTf exists as its “neutral ion pair” in water ($[\text{DMF}]^+[\text{OTf}]^- \rightarrow \text{DMF and HOTf}$), a phenomenon known for other formamidium ionic liquids.^{31,32} The presence of the superacid HOTf explains the unusually low initial pH of 2 M ZnSO_4 -50 mM DOTf ($\text{pH} \approx 1.6$). Similar excellent Zn-plating/stripping performance was observed using 2 M ZnSO_4 -50 mM DMF-50 mM HOTf as the electrolyte, (Figure S12), further corroborating the existence of neutral ion pairs in aqueous solution.

^1H - ^{13}C heteronuclear single quantum coherence (HSQC) experiments were carried out to further characterize the additional peaks obtained in the modified electrolyte. As shown in Figure 3B, three different protons (a, b, and c) correlate with three carbons at around $\delta = 164.8$, 36.9, and 31.4 ppm, characteristic of DMF. The highest frequency proton a' ($\delta = 7.83$ ppm) shows a distinguishable correlation with the carbon at $\delta = 166.1$ ppm. These results point at the possible presence of formic acid, which was confirmed through spiking the solution with its excess (Figure S13). The formation of formic acid from DMF (a result of acid hydrolysis reaction)³³ helped us to identify the proton d ($\delta = 2.28$ ppm) as protonated dimethylamine. While characteristic ion solvent coordination results in a more de-shielded solvent environment,³⁴ our results (Figure S14) depict the opposite behavior. All the DMF peaks are significantly shifted to lower frequency indicative of possible formation of poly-metal clusters in solution that could shield the solvent environment.³⁵ Identifying the exact molecular speciation is beyond the scope of this work and will be part of a future study. These complex molecular species along with the “amine” products obtained through hydrolysis could potentially assist

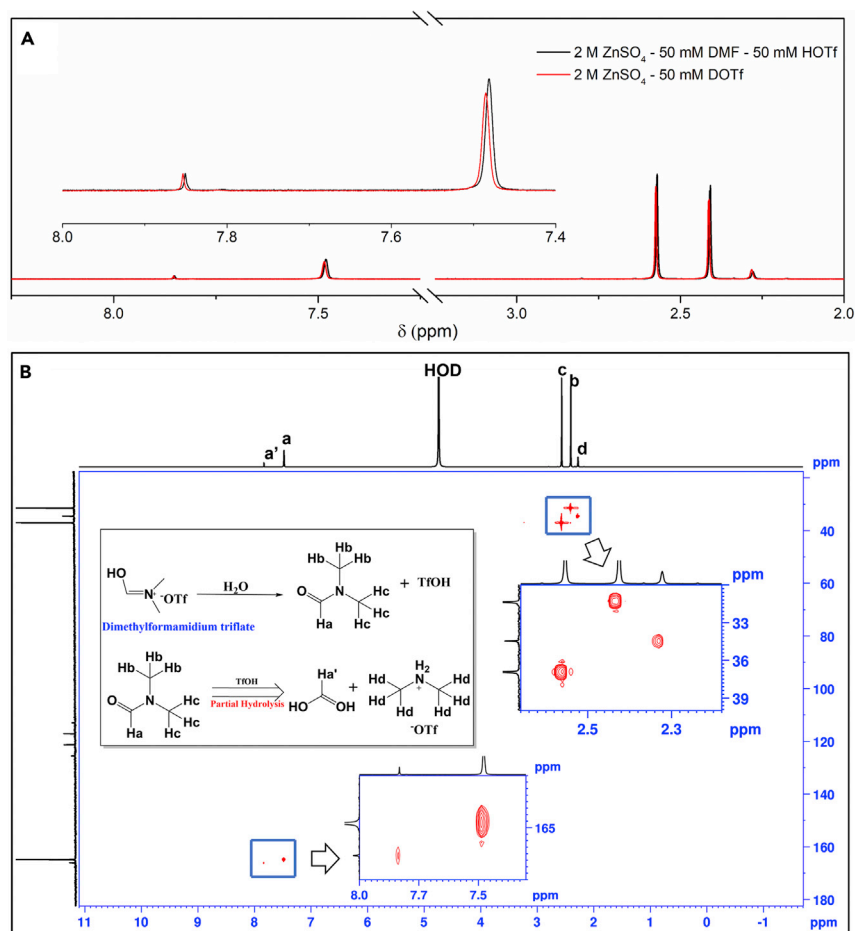


Figure 3. NMR studies for the 2 M ZnSO₄-50 mM DOTf electrolyte

(A) Comparison of the ¹H-NMR spectra of 2 M ZnSO₄ with 50 mM DOTf and 50 mM DMF-HOTf. (B) The HSQC 2D spectrum of the DOTf modified electrolyte. The ¹H and ¹³C patterns are shown on the horizontal axis and the vertical axis, respectively.

the formation of a Zn²⁺-conducting SEI layer on the Zn surface during the electrochemical process.

To delve more into the nature of the SEI, XPS was carried out on the topmost surface of the plated Zn, and at a depth of ~2 and ~10 nm underlying the surface by removing the upper material via 2 and 10 min Ar⁺ sputtering, respectively. The XPS spectra of plated Zn in 2 M ZnSO₄-50 mM DOTf are shown in Figure 4A. All binding energies were calibrated by using the C 1s peak (at 284.8 eV) as the reference. The N 1s spectra shows a stable amine-related component (at 399.4 eV) at different depths. This amine signal—which is absent in the additive-free electrolyte—can be ascribed to either dimethylamine bonded to Zn²⁺, or trimethylamine produced by the electrochemical reduction of DMF.³⁶ The O 1s spectrum of the topmost surface exhibits two major components: SO₄²⁻ (at 532.1 eV, 91%) and ZnO/Zn(OH)₂ (at 530.4 eV, 9%). The sulfate signal can be ascribed to the precipitation of Zn salt in the electrolyte.^{26,37} The S 2p spectrum also confirms the existence of ZnSO₄ (at 169.4 eV, 92%) along with minor ZnS species (at 162.1 eV, 8%). At a depth of ~2 nm (sputtering for 2 min) the ZnSO₄ content reduces to ~77% in both O 1s and S 2p spectra. Meanwhile, both ZnO/Zn(OH)₂ and ZnS fractions increase to

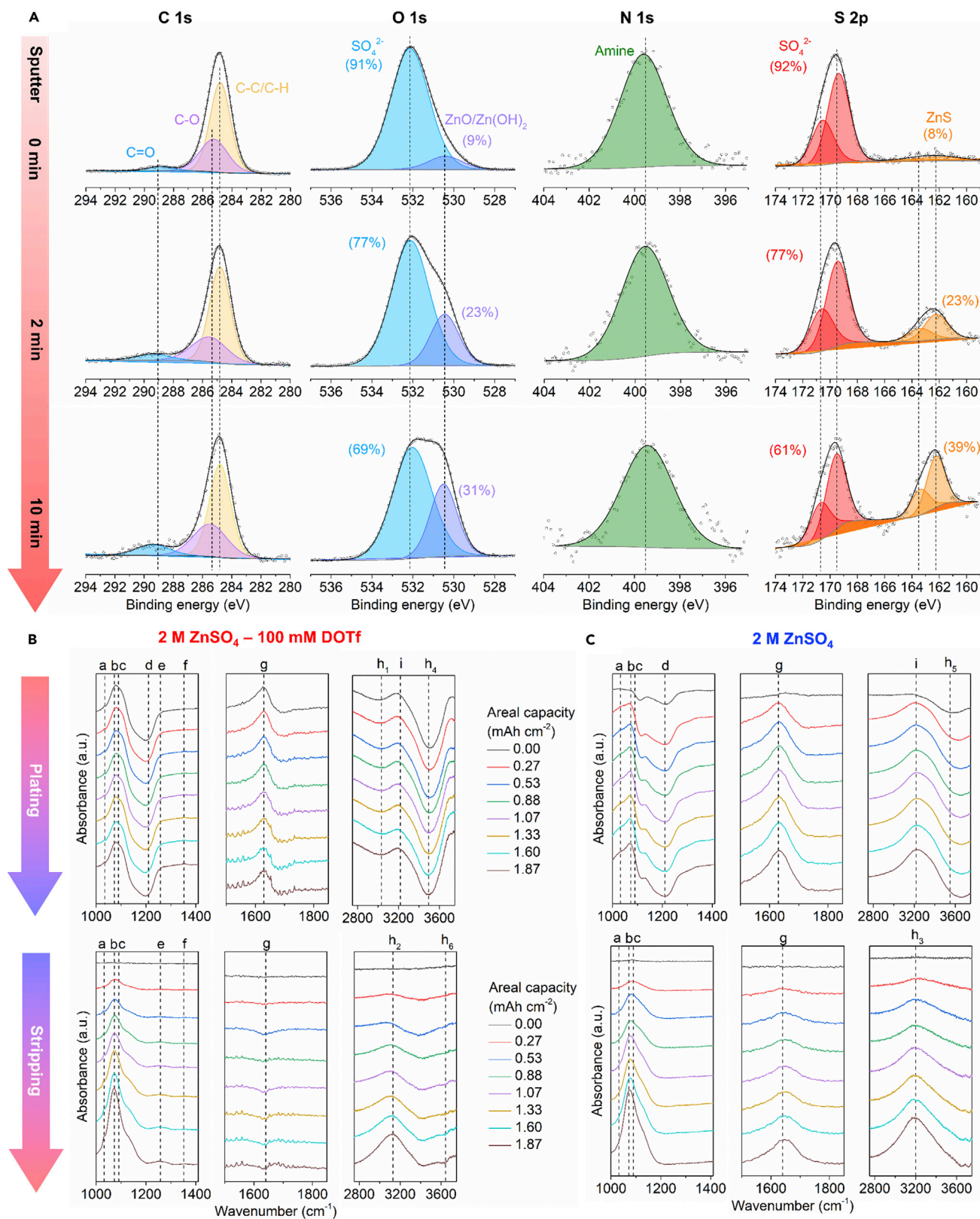


Figure 4. Interfacial studies of the Zn/electrolyte interface in different electrolytes

(A) XPS spectra for 1st plated Zn on Cu in 2 M ZnSO₄-50 mM DOTf at 4 mA cm⁻² and 4 mAh cm⁻².

(B and C) *Operando* SEIRAS of both plating and stripping process on Au in (B) 2 M ZnSO₄-100 mM DOTf and (C) 2 M ZnSO₄ at 4 mA cm⁻² for 30 min (peak assignments shown in Table S2).

~23%. Upon further sputtering to a depth of ~10 nm, the ZnSO₄ signal is further reduced in intensity while both ZnO/Zn(OH)₂ and ZnS increase to reach over 30% of their respective (O 1s, S 2p) signal. The ZnS component probably originates from the reduction of SO₄²⁻ rather than the OTf⁻ anion owing to the lack of fluorine-related species in the F 1s XPS spectrum (Figure S15). This conclusion is supported by the lack of a significant ZnS signal in the XPS obtained from deposits grown in a 2 M Zn(ClO₄)₂-50 mM DOTf electrolyte and where poor Zn reversibility is observed (Figure S16). It is likely that SO₄²⁻ is reduced to ZnS by the H₂ formed by the chemical reaction between Zn and HOTf in the initial cycles (Figure S17), as ZnS is also observed on Zn metal soaked in the modified electrolyte (Figure S18). In addition, no ZnS is observed after soaking Zn in 50 mM DOTf without ZnSO₄ and for electrochemically deposited Zn in the baseline electrolyte, further confirming that SO₄²⁻ anion and the chemically formed H₂ are key to the formation of ZnS (Figure S19).

X-ray diffraction (XRD) analysis reveals that no deleterious LDH forms on plated Zn in the modified electrolyte, whereas well-defined peaks of LDH are observed for the electrode cycled in the baseline electrolyte (Figure S20). The lower potential of HER with the addition of DOTf (Figure S21) also suggests the formation of LDHs is suppressed owing to the lack of a local alkaline environment at the electrode/electrolyte interface. Details of the interfacial chemistry were further studied by SEIRAS (see peak assignments in Table S2 and experimental procedures for details). In these experiments, a reference spectrum was collected immediately prior to Zn plating and immediately prior to Zn stripping and was subtracted from all subsequent spectra. Although free SO₄²⁻ (band d) is displaced from the Au surface during plating owing to the negative surface polarization (Figures 4B and 4C), significant amounts of SO₄²⁻ association with Zn²⁺ leads to the presence of ZnSO₄ (band b) at the interface in both electrolytes. However, in the presence of DOTf, the local acidic environment limits the formation of LDH, possibly allowing for the formation of other products such as ZnS. This acidity also results in lower peak intensities of bands a, c, and i that correspond to Zn(OH)₂. In addition, changes to the OTf⁻ (band e) presence at the interface were observed. While peak e is obscured by the larger peak d during plating, there is a negative-going shoulder of peak d at band e, indicating that OTf⁻ is likely leaving the interface. During stripping, there is a clear positive-going peak e, which indicates that OTf⁻ is drawn to the surface. The presence of formate (band f) at the interface was also identified in both stripping and plating. More importantly, in 2 M ZnSO₄-100 mM DOTf, much less H₂O approaches the interface during plating compared with that in 2 M ZnSO₄, as verified by different intensities of peak g, which corresponds to the H₂O bending mode. In the case of 2 M ZnSO₄-100 mM DOTf during plating, each spectrum following reference subtraction exhibits negative-going peaks h₁ and h₄ assigned to H₂O O-H stretching modes, indicating changing H₂O modes at the interface. For 2 M ZnSO₄ a similar trend is observed with the decrease of peak h₅, although this decrease is somewhat obscured by the stronger positive-going peak i. During stripping, peak h₆ appears at a much higher energy, indicating an interrupted hydrogen bonding network of H₂O at the interface. These changes suggest the presence of a H₂O-deficient interface in the presence of DOTf, thus suppressing LDH formation and enhancing Zn

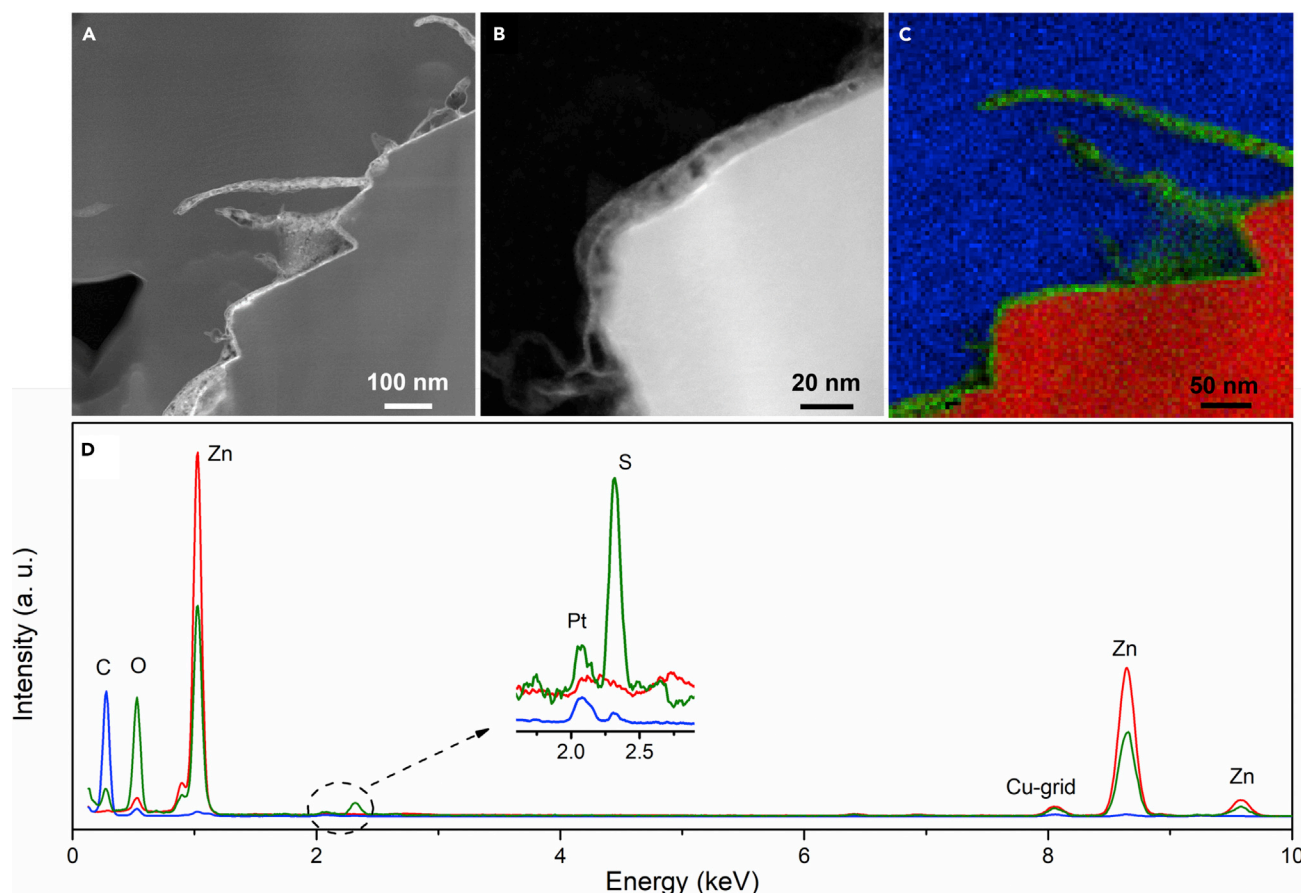


Figure 5. Structural and compositional mapping of the SEI formed in 2 M ZnSO₄-50 mM DOTf

(A and B) Dark-field and HAADF images showing the conformal and porous structure of the SEI.

(C and D) EDS-based spectral imaging of the SEI with the corresponding multivariate spectral components: Zn deposit, red; interphase and nanosheets, green; carbon cap, blue.

reversibility. During stripping, an increase in similar H₂O stretching modes (ν_2 and ν_3) is observed in both electrolytes, which probably results from the increasing concentration of Zn²⁺ at the interface that influences the interaction between H₂O and Zn²⁺.

Structural and chemical mapping of the interphase produced with the DOTf additive was conducted using cryogenic ion milling and cryo-TEM. Low temperature milling reduced the mechanical distortion observed with room temperature milling of the Zn deposit, allowing for more uniform, thinner Zn cross-sections and higher fidelity imaging. A representative dark-field image at the surface of a first cycle Zn deposit (4 mA cm⁻², 4 mAh cm⁻² as in Figure 2B) formed with DOTf is shown in Figure 5A. The interphase appears as a bright contrast, conformal layer at the electrolyte exposed facets of the Zn grains and underneath a nanosheet structure, viewed edge on here, forms with the DOTf additive (Figure 2B). We find the interphase is a porous layer ranging in thickness from 10 nm to less than 5 nm, where pores are seen as dark regions in the high-angle annular dark-field (HAADF) image of Figure 5B. We propose these nanopores are created *in situ* by the initiation of superacid (HOTf) reduction at the Zn surface to release minute quantities of H₂ as described above, that also reduces SO₄²⁻ to form ZnS. Nanoporosity is critical as it

provides the cation access to a sufficiently thin portion of the nanostructured interphase enabling cation transport, thus producing the dense Zn deposits shown to form in the presence of DOTf (Figure 2D). The interphase appears compositionally similar to the nanosheets as a mixture of ZnO and ZnS, with as much as 5% sulfide present, as determined by the multivariate analysis of energy-dispersive X-ray spectroscopic (EDS) imaging data (Figures 5C and 5D).³⁸ The presence of oxide in the interphase is confirmed from lattice spacings characteristic of wurtzite ZnO observed in high-resolution scanning transmission electron microscopy (STEM). We are not able to similarly differentiate the chemical state of S using diffraction or electron energy loss spectroscopy (EELS) due to low concentration and possibly a lack of crystallinity of the ZnS phase. Regardless, the presence of S within the interphase is consistent with the XPS derived composition of the interphase highlighting the DOTf additive's role in forming a nanometric, porous SEI responsible for facilitating Zn cation transport.

In short, these results clearly demonstrate that a uniform nanostructured SEI forms on the Zn surface in 2 M ZnSO₄-50 mM DOTf electrolyte owing to the unusual surface chemistry. Reduced water activity can be anticipated within this SEI layer as result of altered Zn²⁺ solvation structures in these pores, as observed in artificial porous protective layers on Zn anodes (for example, zeolitic membranes³⁹). Such a porous SEI layer allows Zn²⁺ to diffuse but prevents the formation of LDHs on the Zn anode and effectively enhances the reversibility and stability of Zn plating/stripping.

Nanoscale Zn nucleation and growth

The aforementioned SEI also manipulates the initial Zn nucleation and subsequent growth behavior, leading to dendrite-free Zn electrodeposition. The process of nucleation and growth is correlated with the evolution of the voltage profile, which allows us to distinguish the overpotentials related to these processes during Zn electrodeposition. Figure 6A shows the characteristic voltage profiles of the first Zn-plating process in different electrolytes. In the baseline electrolyte, a sharp nucleation overpotential (η) is observed of about 132 mV, whereas in 2 M ZnSO₄-50 mM DOTf, η is not at all well defined and is slightly larger (153 mV). This is ascribed to higher impedance at the DOTf-Zn interface due to the presence of the SEI layer (Figure 5) that promotes homogeneous deposition of more, smaller, and denser Zn nuclei.⁴⁰ The SEM image of plated Zn with a capacity of 0.1 mAh cm⁻² further confirms the uniform and nanoscale Zn deposits in the modified electrolyte, whereas sparse needle-shape Zn nuclei about ~2 μ m in length are observed in 2 M ZnSO₄ (Figures 6B, 6H, and S22). In addition, the difference in the growth overpotential with respect to the nucleation overpotential ($\Delta\eta$) is 12 mV in the DOTf electrolyte, which is approximately one quarter of that without DOTf. Thus, it is more favorable for Zn to deposit as smaller crystals than large dendrites.^{40,41} The morphological evolution of Zn growth was further investigated by *ex situ* SEM. Fixed amounts of Zn from 0.1 to 4.0 mAh cm⁻² were galvanostatically deposited on Cu at 4 mA cm⁻². In 2 M ZnSO₄-50 mM DOTf, nanoscale Zn nuclei grows as small, ~250 nm crystallites, which merge into dense and compact Zn deposition layers (Figures 6B-6G), while random Zn dendrites gradually form in the baseline electrolyte (Figures 6H-6M).

The proposed effect of the SEI on Zn nucleation and growth is shown in Scheme 1. In the baseline electrolyte, porous LDH deposits on the Zn surface owing to the progressive HER and consequently local alkaline environment during Zn electrodeposition (Scheme 1A). Such an inhomogeneous surface leads to a large energy difference between the Gibbs energy for Zn nucleation ($\Delta G_{\text{nucleation}}$) and growth (ΔG_{growth}).

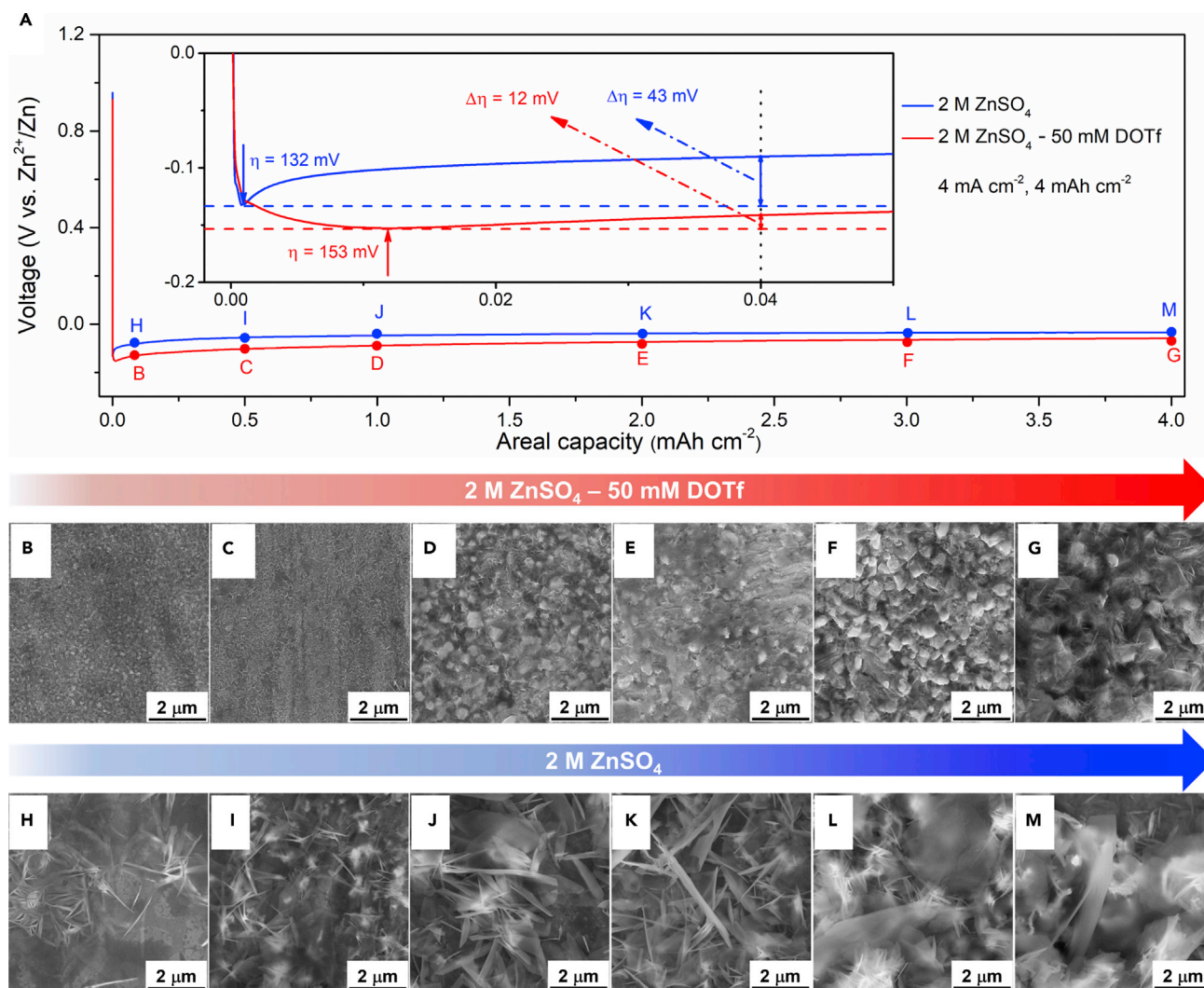


Figure 6. Nanoscale nucleation and growth of plated Zn

(A) The first plating voltage profiles of Zn||Cu cells at 4 mA cm^{-2} and 4 mAh cm^{-2} with a magnified view of initial nucleation process. (B–M) Ex situ SEM images of plated Zn on Cu in $2 \text{ M ZnSO}_4\text{-}50 \text{ mM DOTf}$ (B–G) and 2 M ZnSO_4 (H–M) after: (B and H) 0.1 mAh cm^{-2} , (C and I) 0.5 mAh cm^{-2} , (D and J) 1.0 mAh cm^{-2} , (E and K) 2.0 mAh cm^{-2} , (F and L) 3.0 mAh cm^{-2} , and (G and M) 4.0 mAh cm^{-2} .

Therefore, Zn^{2+} prefers to deposit on the existing Zn nuclei to form large Zn flakes or dendrites. In contrast, the homogeneous nanostructured SEI in the modified electrolyte raises the ΔG_{growth} to the same level of $\Delta G_{\text{nucleation}}$ (Scheme 1B). This significantly reduces the size of the Zn crystallites and results in uniform dendrite-free Zn deposition.

Electrochemistry of $\text{Zn}||\text{Zn}_{0.25}\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$ full cells

The electrochemistry of $2 \text{ M ZnSO}_4\text{-}50 \text{ mM DOTf}$ electrolyte was tested in full cells, (Figure 7) using $\text{Zn}_{0.25}\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$ as a representative positive electrode (Figures S23 and S24) to examine compatibility with the stabilized Zn anode.⁵ The rate performance of $\text{Zn}||\text{Zn}_{0.25}\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$ full cells in $2 \text{ M ZnSO}_4\text{-}50 \text{ mM DOTf}$ electrolyte shows a specific discharge capacity of 326 mAh g^{-1} at 0.3 A g^{-1} (1C rate) and 217 mAh g^{-1} at 2.4 A g^{-1} (Figures 7A and 7B), which is slightly higher than that in the absence of DOTf. This shows good compatibility between

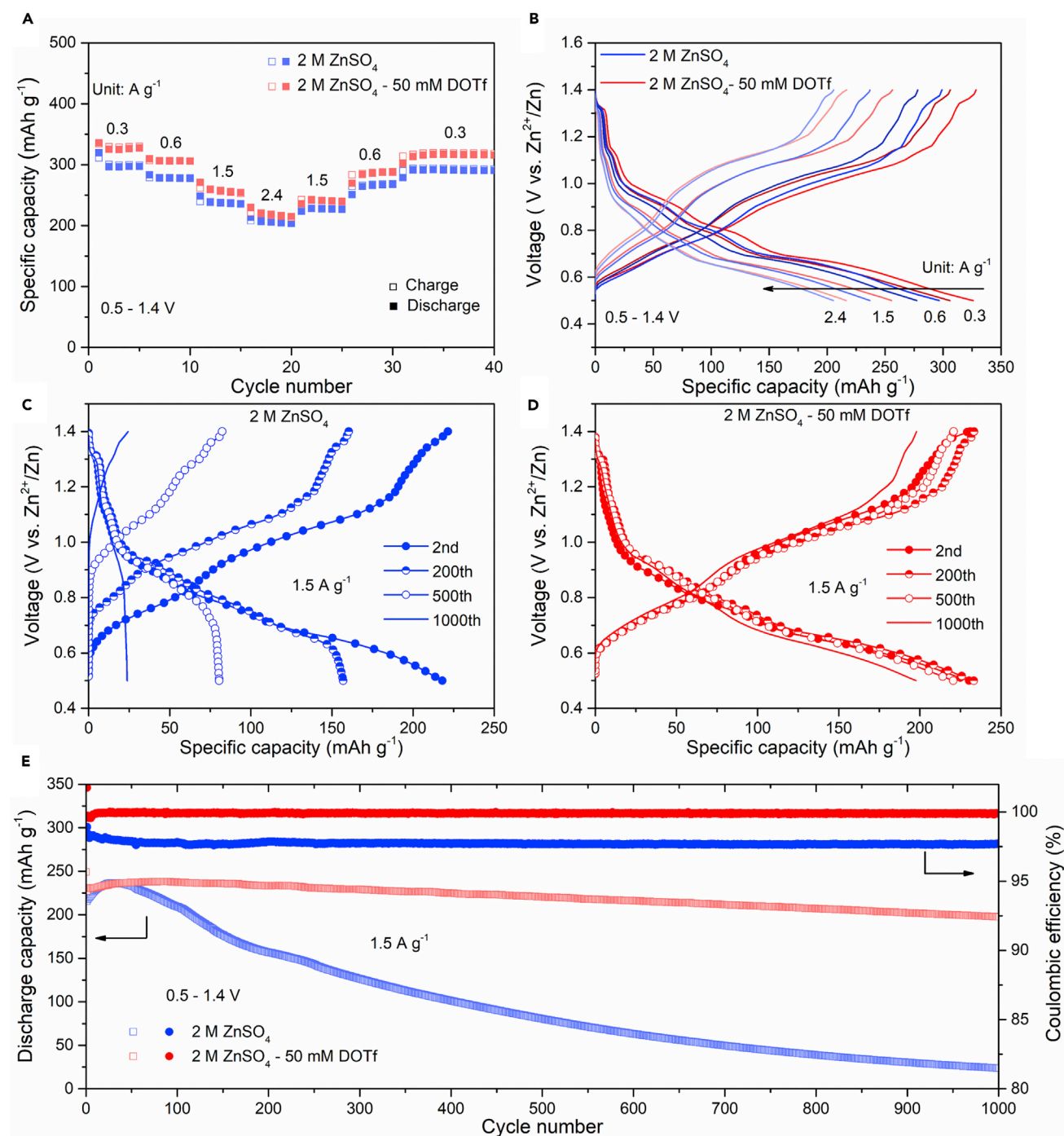


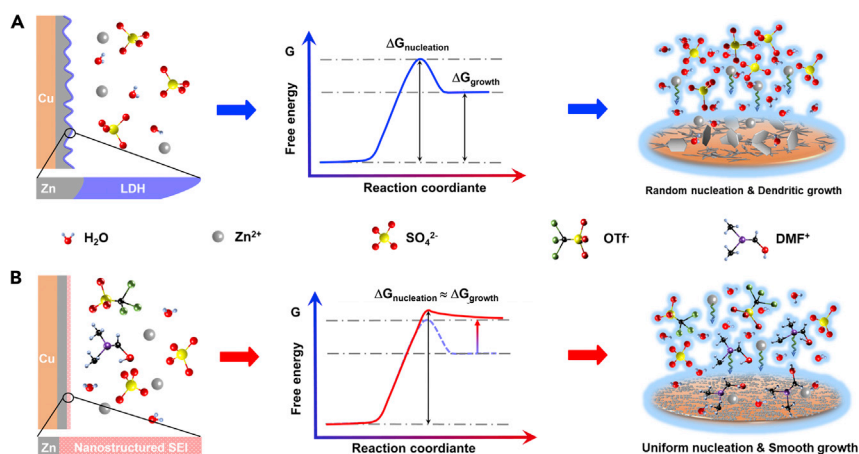
Figure 7. Electrochemistry of $\text{Zn}||\text{Zn}_{0.25}\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$ full cells between 0.5 and 1.4 V

(A and B) Rate performance of full cells in different electrolytes: (A) specific capacity as a function of cycle numbers and (B) corresponding charge-discharge curves at different current densities.

(C and D) The voltage profiles of full cells (C) in 2 M ZnSO_4 and (D) 2 M ZnSO_4 -50 mM DOTf at 5 C ($N:P \approx 2.5$).

(E) Long-term stability of full cells in different electrolytes at 1.5 A g^{-1} ($N:P \approx 2.5$). The Zn anode was deposited on Cu foil.

$\text{Zn}_{0.25}\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$ cathode and the modified electrolyte, yielding an average discharge voltage of $\sim 0.7 \text{ V}$ and energy densities of $228 \text{ Wh kg}_{\text{ZVO}}^{-1}$ or $714 \text{ Wh L}_{\text{ZVO}}^{-1}$ (based on the theoretical density of $\text{Zn}_{0.25}\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$ ($n=1$) of 3.13 g cm^{-3}).



Scheme 1. Illustration of SEI formation and its role in tuning Zn nucleation and growth

(A) 2 M ZnSO_4 electrolyte.

(B) 2 M ZnSO_4 -50 mM DOTf electrolyte.

cm^{-3}) at 0.3 A g^{-1} . Long-term Zn reversibility of the full cell was further evaluated in both electrolytes at a current density of 1.5 A g^{-1} (5C rate) with a controlled anode/cathode (N:P) capacity ratio of 2.5 (Figures 7C–7E). The estimated volumetric energy density in the modified electrolyte based on both cathode and anode is $\sim 400 \text{ Wh L}^{-1}$ (the thickness of deposited Zn is estimated as $\sim 5 \mu\text{m}$ based on Figure 2D). In the baseline electrolyte, a sharp voltage drop at the end of discharge gradually appears with dramatic capacity loss with continued cycling (Figure 7C), which is mainly ascribed to the failure of the Zn anode. However, in the modified electrolyte, the high Zn reversibility enables stable charge/discharge of the $\text{Zn}||\text{Zn}_{0.25}\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$ cell (Figure 7D) at the controlled N:P ratio. As shown in Figure 7E, in the presence of the DOTf additive, this full cell still maintains $\sim 197.6 \text{ mAh g}^{-1}$ after 1,000 cycles, exhibiting a capacity retention of $\sim 83\%$ with an average CE of 99.9% (Figure R4 and R5). In contrast, the capacity in 2 M ZnSO_4 electrolyte quickly drops to $\sim 24 \text{ mAh g}^{-1}$ owing to the poor Zn reversibility with a lower average CE of 97.7%.

Proton intercalation has been reported to be deleterious to the long-term cycling of various metal oxides in AZMBs as it results in the formation of LDHs.^{1,42–45} Similar to the anode side, LDHs also deposit on the cathode owing to the local alkaline environment after protons intercalate into the cathode host as shown schematically in Figure 8A. However, the thin, flaky LDHs tend to spontaneously detach from the cathode side, causing irreversible capacity loss and poor long-term cycling stability.^{1,43,44} XRD analysis on $\text{Zn}_{0.25}\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$ cathodes collected at different discharge states shows that LDH mainly forms in the low voltage range from 0.7 to 0.5 V (Figure S25). The discharge curves of $\text{Zn}_{0.25}\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$ cathodes are shown in Figure 8A. Up to the initial discharge state of 0.7 V, Zn^{2+} gradually intercalates into the lattice with a small portion of proton intercalation and concomitant formation of LDH on the cathode surface. Below 0.7 V, proton intercalation and LDH formation dominate the discharge process and account for the most of electrochemical capacity, consistent with previous reports.^{5,46} This allows us to further enhance the stability of $\text{Zn}||\text{Zn}_{0.25}\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$ full cells by modifying the lower cutoff voltage from 0.5 to 0.7 V based on a similar strategy.⁴⁷ Figure 8B shows that this full cell still maintains $\sim 84\%$ of its initial highest capacity after even 2,000 cycles at the same Zn capacity with an average CE of 99.3% (N:P ratio increases to ~ 5.2 owing to the

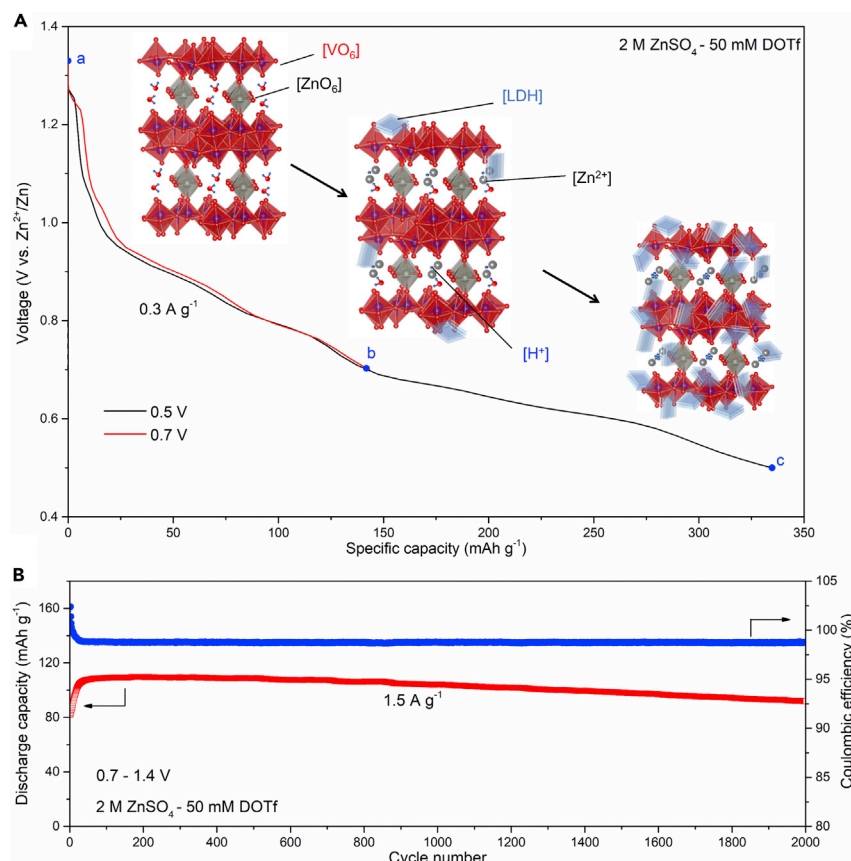


Figure 8. H⁺/Zn²⁺ Intercalation of Zn||Zn_{0.25}V₂O₅·nH₂O full cells in 2 M ZnSO₄-50 mM DOTf

(A) The first voltage profiles of full cells at different discharge states and corresponding schematic illustration of H⁺/Zn²⁺ intercalation into the lattice (a, b, and c indicate the open circuit voltage, 0.7 and 0.5 V).

(B) Long-term stability of the full cell between 0.7 and 1.4 V at 1.5 A g⁻¹ (N:P ≈ 5.2, the areal capacity of deposited Zn on Cu is the same as that in 0.5–1.4 V).

lowered cathode capacity, Figure R6 and R7), which is twice the cycle life at 0.5 V. We note that this promising stability is obtained at a moderate current density of 5C, while many literature reports utilize very high rates (>10C) to favor stability.^{9,20} At a 2C rate (0.6 A g⁻¹), ZVO also exhibits good cycling stability at the lower cutoff voltage of 0.7 V, with a capacity retention of ~90% after 500 cycles (Figure S26). The enhanced cycling stability can be attributed to the alleviated proton intercalation, where the capacity loss owing to the detachment of LDH particles is significantly reduced.

Conclusions

In this work, we introduce DOTf as a novel, low-concentration electrolyte additive for aqueous Zn metal batteries. The use of this electrolyte leads to dendrite-free and highly reversible Zn plating/stripping for more than 3,500 h in Zn||Cu cells. More importantly, ~100% average CE of Zn plating/stripping is achieved under a practical cycling condition of a high current density of 4 mA cm⁻² and a high areal capacity of 4 mAh cm⁻² for more than 250 cycles. The superior Zn stability and reversibility are ascribed to robust nanostructured SEI layers leading to the suppression of detrimental LDHs and a nanoscale nucleation-growth process during Zn

electrodeposition. The origin of this Zn^{2+} -conducting SEI is speculated to be a result of the water-assisted dissociation of DOTf into neutral ion pairs. $\text{Zn}||\text{Zn}_{0.25}\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$ full cells (N:P ratio of 2.5 and 5.2) based on this electrolyte exhibit excellent cycling performance with different depth of proton intercalation. Our findings and understanding of this novel electrolyte additive could potentially accelerate the commercialization of practical aqueous Zn metal batteries.

EXPERIMENTAL PROCEDURES

Resource availability

Lead contact

Further information and requests for resources and materials should be directed to and will be fulfilled by the lead contact, Linda F. Nazar (lfnazar@uwaterloo.ca).

Materials availability

This study did not generate new unique materials.

Data and code availability

The datasets generated in this study are available from the lead contact on reasonable request.

Synthesis of $\text{Zn}_{0.25}\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$ nanobelts

$\text{Zn}_{0.25}\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$ nanobelts were synthesized by a previously reported microwave hydrothermal method.⁵ Typically, 2 mmol V_2O_5 (99.6%; Sigma-Aldrich) and 1.3 mmol $\text{Zn}(\text{CH}_3\text{COO})_2 \cdot 2\text{H}_2\text{O}$ (99%; Sigma-Aldrich) were dispersed in 50 mL water/acetone (volume ratio: 15:1) solution. The mixture was transferred to a sealed Teflon vessel and kept at 180°C for 90 min at a heating rate of 18°C min⁻¹ in an Anton Parr microwave (Synthos 300). The final $\text{Zn}_{0.25}\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$ nanobelts were obtained by filtration, washing with deionized water and isopropanol, and drying in a 60°C oven for 24 h.

Electrochemical measurements

The $\text{Zn}_{0.25}\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$ cathodes were prepared by doctor blading the mixture of active materials, conductive carbon Super P and polyvinylidene difluoride (PVDF) in a weight ratio of 7:2:1 in N-methyl-2-pyrrolidone (NMP; anhydrous, 99.5%; Sigma-Aldrich) on carbon paper (thickness: 170 μm). After drying at a 60°C oven overnight, the electrodes were punched into small coins (diameter: 10 mm). The mass loading of $\text{Zn}_{0.25}\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$ was $4.0 \pm 0.5 \text{ mg cm}^{-2}$. The full cells were assembled in PFA-based Swagelok-type cells, using the commercial Whatman glass fiber as separators and Cu foil (thickness: 9 μm ; diameter: 11 mm) with pre-deposited Zn (about 3 mAh cm⁻² for both 0.5 and 0.7 V voltage cutoffs) as the counter and reference electrodes with controlled N:P ratios. The $\text{Zn}||\text{Cu}$ asymmetric and $\text{Zn}||\text{Zn}$ symmetric cells were assembled with Zn metal foil (99.9%; Sigma-Aldrich; thickness: 250 μm ; diameter: 10 mm) as the electrodes. The amount of electrolyte in each cell is 100 μL . The galvanostatic cycling were carried out on a VMP3 potentiostat/galvanostat station (Bio-logic) at room temperature. Linear sweep voltammetry (LSV) was collected based on a three-electrode cell with Zn as reference electrode, Ti as both working and counter electrodes. Additional explanation of cell assembly and data collecting can be seen in [Data S1](#).

2 M ZnSO_4 was prepared by dissolving 10 mmol $\text{ZnSO}_4 \cdot \text{H}_2\text{O}$ (> 99%; Sigma-Aldrich) into 5 mL deionized water. 2 M ZnSO_4 -50 mM DOTf electrolyte was obtained by adding 0.25 mmol dimethylformamidium trifluoromethanesulfonate (97%; Sigma-Aldrich) into 5 mL 2 M ZnSO_4 .

Materials characterization

Powder XRD studies were carried out on an X-ray diffractometer using Cu-K α radiation (PANalytical Empyrean) equipped with a PIXcel bidimensional detector. SEM images and EDX were collected by a Zeiss ultra field emission SEM instrument equipped with an EDX detector. XPS analysis was conducted on a Thermo Scientific K-Alpha XPS instrument. The binding energies were calibrated using the C 1s peak at 284.8 eV. All spectra were fitted with Gaussian-Lorentzian functions and a Shirley-type background using CasaXPS software. The S 2p doublet peaks were fit using equal full width at half maximum (FWHM), with a 2:1 area ratio and splitting of 1.16 eV between the S 2p 3/2 and S 2p 1/2. The spot size for photoelectron collection was a 400 μm $\times$ 200 μm ellipse, and the grid for sputtering was a 1,600 μm $\times$ 800 μm rectangle, with an Ar⁺ etching rate estimated to be $\sim 1.0 \text{ nm min}^{-1}$. NMR experiments were carried out on a Bruker Avance 300 MHz instrument at ambient temperature. All samples were dissolved in D₂O which served as an internal lock and were loaded into 5 mm tubes (800 μL). ¹H NMR experiments were carried at 300 MHz and ¹³C NMR experiments were carried at 75 MHz. HSQC experiments were carried out to correlate the chemical shift of protons (F2 axis) to the ¹³C chemical shift (F1 axis) of their directly attached carbons. All electrodes that were characterized were collected from the cell, rinsed thoroughly with deionized water and ethanol, and dried at 80°C *in vacuo* overnight. Attenuated total reflectance (ATR)-SEIRAS was carried out by using a Nicolet Magna-IR 550 spectrometer with a home-built spectro-electrochemical cell that was previously described.⁴⁸ *In situ* spectra were collected using the roughened 20 nm Au as the working electrode and a Zn wire as counter and reference electrode. Zn was galvanostatically plated 4 mA cm⁻² for 30 min for both electrolytes and stripped at 4 mA cm⁻² for 30 min for the 2 M ZnSO₄-100 mM DOTf electrolyte and 28 min for the 2 M ZnSO₄ electrolyte. The reference spectrum was collected immediately prior to Zn plating and immediately prior to Zn stripping, which was subtracted for all subsequent spectra. A Helios NanoLab 600 FIB/SEM was used to deposit a protective, infiltrating carbon capping layer onto the Zn deposit surface prior to cooling for cryogenic FIB milling, lift-out, and final thinning in a Scios 2 FIB/SEM. The resulting cross-section lamella was stored in liquid nitrogen and loaded into a Gatan 626 cryo holder for cryogenic-STEM, EDS, EELS characterization in an FEI Titan G2 operated at 200 kV. The sulfur K intensity was calibrated using a ZnS standard.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.joule.2022.04.017>.

ACKNOWLEDGMENTS

This work was financially supported by the Joint Centre for Energy Storage Research, an Energy Innovation Hub funded by the US Department of Energy, Office of Science, Basic Energy Sciences. This work was performed, in part, at the Center for Integrated Nanotechnologies (CINT), an Office of Science User Facility operated for the US Department of Energy (DOE), Office of Science. Sandia National Laboratories is a multimission laboratory managed and operated by National Technology & Engineering Solutions of Sandia, LLC, a wholly owned subsidiary of Honeywell International, Inc., for the US DOE's National Nuclear Security Administration under contract DE-NA0003525. This paper describes objective technical results and analysis. Any subjective views or opinions that might be expressed in the paper do not necessarily represent the views of the US Department of Energy or the United States Government. We would like to thank L. Zhou (University of Waterloo) for

helpful discussions and P. Brodersen (University of Toronto) for carrying out the XPS experiments. L.F.N. also acknowledges NSERC for platform support through the Discovery Grant and Canada Research Chair programs.

AUTHOR CONTRIBUTIONS

C.L. and L.F.N. designed this study. C.L. synthesized the materials and carried out the characterization and all the electrochemical measurements. A.S. performed NMR measurement. A.G.H. and A.A.G. carried out the SEIRAS. C.Y.K. helped with XPS experiments and data analysis. D.M.L., P.G.K., and K.R.Z. conducted TEM measurements. C.L. and L.F.N. wrote the manuscript with contributions from A.G.H., A.A.G., D.M.L., and K.R.Z.

DECLARATION OF INTERESTS

The authors declare no competing interests.

Received: December 22, 2021

Revised: January 25, 2022

Accepted: April 20, 2022

Published: May 18, 2022

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